

ARITRA KARMAKAR

FULL STACK DEVELOPER

CONTACT

+919749185898
aritra.karmakar9749@gmail.com
<https://portfolioofaritra.vercel.app>
Bengaluru, IN

SKILLS

PROGRAMMING LANGUAGES

JavaScript Python

TypeScript

FRAMEWORKS & LIBRARIES

React Next.js
Node Express.js
Redux Django
Tailwind CSS Flask
NumPy Pandas
React Native TanStack Query

CLOUD TECHNOLOGIES

AWS Kubernetes
CI / CD Docker
Serverless Vercel

DATABASES & TOOLS

PostgreSQL MySQL
MongoDB RESTful API
GraphQL Git
Postman Nginx
Prisma Redis
Cloudflare Worker Cloudflare KV
QGIS Linux

PROFILE

A versatile and highly motivated Full Stack Developer with over 4 years of experience, including 2+ years of professional work experience. Skilled in React, Next.js, Python, DevOps, and data analysis, with a strong foundation in designing and deploying scalable applications. Experienced in leading end-to-end software development projects and collaborating with cross-functional teams to deliver high-quality products on time.

EDUCATION

2019 - 2023

B.E.
Information Science & Engineering

Visvesvaraya Technological University,
Bengaluru, India

WORK EXPERIENCE

Full Stack Developer

Dec 2023 - Present (1yr 2m)

CISTUP, Indian Institute of Science, Bengaluru

- Collaborated with IISc researchers and BOSCH developers on the "City-Scale Travel Demand Model Development" project, funded by DULT, Government of Karnataka.
- Worked with cross-functional teams to align project execution with organizational goals.
- Developed a large-scale full-stack digital survey web application using **React**, **Python Flask**, and **PostgreSQL**.
- Utilized **useReducer** and **useContext** React hooks with local storage for efficient state management.
- Enhanced server API stability and security by implementing a **proxy** between servers.
- Managed production deployment with **Nginx**, employing **reverse proxy** and **load balancing** techniques.
- Facilitated accurate surveying of 10,000 households in Bangalore, ensuring precise and **reliable data collection** using the Digital Survey Tool.
- Designed and developed scripts for presentable **Exploratory Data Analysis (EDA)** and **Data Sanity Checks** using **Pandas**, **NumPy** and **forum**, ensuring data integrity and quality.
- Created a data clustering algorithm to identify fake surveys by enumerators.
- Led **testing** and **bug-fixing** for the Aggregate Travel Demand Model QGIS plugin, generating test cases and creative scenarios.
- Reduced web-based output visualization dashboard load time by 80% in a QGIS plugin integrated with **Flask**, utilizing **DuckDB's SQL** support and in-memory processing to efficiently manage large CSV files.
- Developed and integrated key modules for the Disaggregate Travel Demand Model QGIS plugin using **Python**.
- Applied reverse engineering and debugging techniques on the open-source project **Activity Sim** to expedite the development of the Disaggregate Travel Demand Model QGIS plugin.
- Implemented **YAML-driven** configuration logic for large-scale data processing in the Disaggregate Travel Demand Model QGIS Plugin.
- Designed and implemented a **test environment** for validating sub-models within the pipeline, ensuring seamless **integration testing**.

EXPERTISE

- Software Development Life Cycle (SDLC)
- Agile Project Management
- Debugging and Troubleshooting
- Version Control (Git, GitHub, GitLab)
- Code Optimization and Refactoring
- Software Design Patterns
- Continuous Integration/Continuous Deployment (CI/CD)
- Test-Driven Development (TDD)
- Requirement Analysis and Documentation

ACHIEVEMENTS

SMART INDIA HACKATHON 2022

CUTM, Bhubaneswar 24th - 29th August 2022

- **Grand Finale Selection:** Selected for the **Grand Finale** of **Smart India Hackathon (SIH) 2022**, showcasing exceptional problem-solving and innovation skills.
- **Project Work:** Developed and delivered a **MATLAB** simulation for the project titled '**Design an Indigenous Monitoring Communication Receiver**', addressing the specific requirements of the problem statement with precision and technical proficiency.

CERTIFICATIONS

 SOFTWARE ENGINEER 

 FRONTEND DEVELOPER 
(REACT)

QUALITIES

- Passionate & Self-directed
- Quick-witted Learner
- Meticulous & Pragmatic
- Innovative & Thoughtful

LINKS



Product Engineering Intern

OSlash, Bengaluru

Dec 2023 - Jan 2024 (1m)

- Collaborated with the Snippets (Product) team to maintain and enhance the OSlash product Dashboard.
- Enhanced web performance by 20% through the elimination of prop drilling and implementation of server components in **Next.js**.
- Resolved 2 major and 1 minor issues in the Product Dashboard using **React** and **TypeScript**, enhancing user experience and functionality.
- Leveraged **TypeScript** to ensure type safety and improve code maintainability.
- Implemented **serverless** functions using **Cloudflare Workers** to enhance application scalability, reduce latency, and improve performance at the edge.

Full Stack Developer Intern

Edureka, Bengaluru

Sept 2021 - Mar 2022 (7m)

- Collaborated with the R&D (Web Services) team to deliver a client-facing, fully responsive e-commerce platform using the MERN stack, tailored to meet real-time business requirements.
- Designed and implemented key features inspired by Zomato, including seamless payment integration through **Paytm JS Checkout** and secure **user authentication**.
- Worked on Edureka's Learning Management System (LMS) to resolve support tickets, address real-time bugs, and ensure system stability for end-users.
- Utilized **AWS services** such as **EC2**, **ECS**, **S3**, and **CloudWatch** to enhance scalability, storage, and monitoring of the LMS infrastructure.
- Oversaw **database migrations** and implemented workflow improvements aligned with DevOps best practices, ensuring seamless deployment and reduced downtime.

BESPOKE PROJECTS

Cloud IDE

- Developed a cloud-based IDE with real-time collaborative features, utilizing **React** for the frontend and **Monaco Editor** for code editing.
- Implemented state management using **useContext** and **useReducer**, combined with **local storage** to ensure a seamless user experience.
- Established a robust client-server bridge with **Node.js** and **Express APIs**, enabling efficient communication and file management.
- Integrated **Socket.io** to synchronize file changes, system updates, and terminal commands in real-time, ensuring immediate reflection of user actions for a seamless experience.
- Utilized **Chokidar** to monitor and dynamically update the file tree structure, enhancing responsiveness and user experience within the user's **isolated** environment.
- **Tech-Stack:** JavaScript, React, Monaco Editor, Node.js, Express, Socket.io, JWT, Chokidar, xTerm, Tailwind CSS, MongoDB

COLLABORATIVE, REAL-TIME WHITEBOARD

- Developed a real-time collaborative whiteboard, inspired by Miro, with tools for **adding shapes**, **sticky notes**, and **drawings**, using **Next.js 14** and **TypeScript** for a robust frontend.
- Implemented a versatile toolbar offering options for text, shapes, sticky notes, and pencil drawings, enhancing user creativity and efficiency.
- Integrated a **layering system** to enable efficient organization and management of elements on the whiteboard.
- Enabled real-time multi-user collaboration using **LiveBlocks**, ensuring seamless updates and interactions for all users.
- Used **Clerk** for authentication, **Convex** for database, and styled with **Tailwind CSS** and **Shadcn UI** for a responsive design.
- **Tech-Stack:** TypeScript, Next.js 14, Clerk, Convex, LiveBlocks, Tailwind CSS.